

Real-Time Time-Dependent Density Functional Theory for Periodic and Spin-Polarized Systems with the Real-Space Multigrids (RMG)[†]

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Abstract

[Abstract: to be written last, after all sections are complete. Approximately 200 words. Should mirror paper 1 abstract structure: (1) what we present, (2) what problem it solves, (3) key methodological contribution, (4) validation/benchmarking, (5) scalability demonstration and outlook.]

1 Introduction

Real-time time-dependent density functional theory (RT-TDDFT) has established itself as one of the primary tools for simulating the electronic response of materials to external electromagnetic perturbations.^{1,2} Its applications span a broad range of phenomena, including optical spectroscopy,³⁻⁷ charge transfer and photovoltaics,^{2,8,9} nanoplasmonics,¹⁰⁻¹² and spin and magnetization dynamics.¹³⁻¹⁵ In an earlier publication, we presented a scalable RT-TDDFT implementation within the Real-space MultiGrid (RMG) code for finite molecular systems, demonstrating simulations of up to 24,000 electrons on massively parallel architectures.¹⁶ The present work extends that implementation to periodic systems and spin-polarized calculations, significantly broadening the range of physically relevant problems accessible to RMG.

The transition from finite to periodic systems in RT-TDDFT requires a corresponding transition in the treatment of the electromagnetic perturbation. In finite systems, the length

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gauge—in which the coupling to an electric field is expressed through the dipole operator $\hat{\mathbf{r}}$ —is natural and convenient. In periodic systems, however, the position operator is ill-defined because it is not translationally invariant, and the dipole moment of an extended system is intrinsically multivalued.^{??} The correct framework for periodic systems is the velocity gauge, in which the electromagnetic interaction is expressed through the vector potential $\mathbf{A}(t)$ and the current operator, which is inherently translationally invariant. The transition between the two gauges is effected by a gauge transformation of the time-dependent Kohn-Sham equations,^{???} and requires careful treatment of the nonlocal part of pseudopotentials.^{???} The primary observable in the velocity gauge is the macroscopic current density $\mathbf{J}(t)$, from which the optical conductivity $\sigma(\omega)$ and dielectric function $\epsilon(\omega)$ are extracted by Fourier analysis.

RT-TDDFT for periodic systems has been implemented in several codes. The **Octopus** code,¹⁷ based on real-space grids, provides a comprehensive periodic RT-TDDFT implementation including both length and velocity gauges and has served as the primary validation reference for the present work. Plane-wave codes including **Qbox/Qball**,^{18,19} **cp2k**,²⁰ and **Exciting**[?] offer periodic RT-TDDFT with natural handling of Born-von Kármán boundary conditions. Atom-centered basis set codes such as **SIESTA**²¹ and **DFTB+**²² provide efficient implementations for systems where localized orbitals are appropriate. The **DFT-FE** code²³ offers an alternative real-space approach based on finite elements. [\[Check: add any missing periodic RT-TDDFT codes before submission. In particular: GPAW, Abinit, Yambo.\]](#)

The treatment of spin-polarized systems within RT-TDDFT introduces an additional degree of freedom: the electronic density is resolved into spin-up and spin-down components, and the time-dependent exchange-correlation potential depends on both. In the collinear approximation, the two spin channels evolve independently subject to a common self-consistent spin-dependent potential, and the optical response carries spin-specific signatures. This capability is essential for magnetic insulators such as CrI_3 , where the spin-up and spin-down optical spectra differ markedly and encode information about the electronic and magnetic structure.^{13,15} The non-collinear spin extension, in which the electronic states are represented as two-component spinors, is beyond the scope of the present work and will be addressed in a future publication.

A distinctive feature of the **RMG** implementation is its scalability. The real-space multi-grid approach assigns different regions of space to different processors, enabling near-linear scaling to hundreds or thousands of GPU-accelerated nodes.^{16,24} This scalability is particularly significant for periodic calculations with spin, where both the k-point sampling and the spin-polarized density matrix increase the computational cost relative to finite systems. To demonstrate this capability, we apply the periodic spin-polarized RT-TDDFT implementation to nitrogen-vacancy (NV) centers in diamond—a prototypical point defect system of substantial interest for quantum sensing and quantum information applications.[?] By systematically increasing the supercell size from [\[X\]](#) to [\[Y\]](#) atoms, we show how the optical response of the defect converges as the interaction between periodic images becomes negligible, identifying the minimum supercell size required for physically meaningful results at realistic defect concentrations.

The remainder of this paper is organized as follows. Section ?? develops the theoretical framework, beginning with the interaction of matter with electromagnetic radiation in the long-wavelength approximation, proceeding through the gauge transformation from length to velocity gauge, the definition of the current operator including nonlocal pseudopoten-

tial contributions, the density matrix propagation scheme, the spin-collinear extension, and the postprocessing route from current to optical conductivity. Section 3 describes the algorithm as implemented in **RMG** and identifies the key approximations and their physical justification. Section 4 presents benchmarking calculations for one-dimensional (H_2 chain), two-dimensional (hexagonal BN), and three-dimensional (Si) periodic systems, followed by spin-polarized calculations for CrI_3 , with comparisons to **Octopus** in all cases. The section concludes with the NV-center in diamond scalability demonstration. The paper concludes with a Summary and Outlook.

2 Theory and Methodology

2.1 Periodic RT-TDDFT and the Velocity Gauge

The starting point for the present periodic RT-TDDFT formulation is the time-dependent Kohn–Sham equation in the presence of an external electromagnetic perturbation,

$$i\hbar \frac{\partial}{\partial t} \Psi_j(\mathbf{r}, t) = \hat{H}(t) \Psi_j(\mathbf{r}, t), \quad (1)$$

where

$$\hat{H}(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + q_e \Phi(\mathbf{r}, t) + \hat{V}_{\text{KS}}[\rho](t). \quad (2)$$

Here $q_e = -e$ is the electron charge, with $e > 0$, m_e is the electron mass, c is the speed of light, and

$$\hat{\mathbf{p}} = -i\hbar \nabla \quad (3)$$

is the canonical momentum operator. Thus, for an electron, the vector potential enters the kinetic momentum as $\hat{\mathbf{p}} + e\mathbf{A}/c$. In the norm-conserving pseudopotential formulation used here, the Kohn–Sham potential is decomposed as

$$\hat{V}_{\text{KS}}[\rho](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{NL}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}[\rho](t), \quad (4)$$

where $\hat{V}_{\text{ion}}^{\text{loc}}$ is the local ionic pseudopotential, $\hat{V}_{\text{NL}}$ is the nonlocal pseudopotential operator, $\hat{V}_{\text{H}}$ is the Hartree potential, and $\hat{V}_{\text{xc}}$ is the exchange-correlation potential.

The physical electric and magnetic fields are obtained from the electromagnetic potentials according to

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \Phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t}, \quad \mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (5)$$

Different choices of Φ and $\mathbf{A}$ related by a gauge transformation describe the same electromagnetic fields. This gauge freedom should be distinguished from the long-wavelength approximation used for optical response. In the present work we consider the homogeneous-field limit, in which the spatial variation of the external field over the simulation cell is neglected. This approximation corresponds to the optical, $\mathbf{q} \rightarrow 0$, limit and leads to vertical transitions at fixed crystal momentum. The $\mathbf{q} \rightarrow 0$ restriction therefore arises from the homogeneous-field approximation, not from the choice of velocity gauge.

For finite systems, a spatially uniform electric field is commonly represented in the length gauge by setting

$$\mathbf{A}_L(\mathbf{r}, t) = 0, \quad \Phi_L(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{E}(t). \quad (6)$$

The corresponding scalar-potential energy for an electron is $q_e \Phi_L(\mathbf{r}, t) = e \mathbf{r} \cdot \mathbf{E}(t)$. This representation is well suited to finite molecular systems, where the time-dependent dipole moment is a well-defined observable. Under periodic boundary conditions, however, the position operator is not compatible with Born–von Karman periodicity: $\mathbf{r}$ and $\mathbf{r} + \mathbf{L}$ represent equivalent points of the crystal but give different values of the scalar potential in Eq. (6). The length-gauge scalar potential is therefore not a periodic operator.

The same homogeneous electric field can be represented in the velocity gauge by a gauge transformation. For a scalar gauge function $\chi(\mathbf{r}, t)$,

$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \nabla \chi(\mathbf{r}, t), \quad \Phi'(\mathbf{r}, t) = \Phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \chi(\mathbf{r}, t)}{\partial t}, \quad (7)$$

and the electronic wavefunction transforms as

$$\Psi'(\mathbf{r}, t) = \exp \left[\frac{ie}{\hbar c} \chi(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t). \quad (8)$$

Starting from the length-gauge potentials in Eq. (6), choose

$$\chi(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{A}(t), \quad \frac{d\mathbf{A}(t)}{dt} = -c\mathbf{E}(t). \quad (9)$$

This gives

$$\Phi_V(\mathbf{r}, t) = 0, \quad \mathbf{A}_V(\mathbf{r}, t) = \mathbf{A}(t), \quad \mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}. \quad (10)$$

For an electron, the corresponding unitary transformation is

$$\hat{U}(t) = \exp \left[-\frac{ie}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right], \quad \Psi_V(\mathbf{r}, t) = \hat{U}(t) \Psi_L(\mathbf{r}, t). \quad (11)$$

Because $\mathbf{A}(t)$ is spatially uniform, $\nabla \times \mathbf{A}(t) = 0$; the homogeneous velocity-gauge field is therefore the gauge-equivalent representation of the electric-dipole field, not a full finite- $\mathbf{q}$ transverse electromagnetic wave. Finite- $\mathbf{q}$ response functions lie outside the homogeneous-field formulation considered here.

This relation also fixes the formal connection between a conventional broadband electric-field kick and the corresponding velocity-gauge vector potential. For an impulsive perturbation,

$$\mathbf{E}(t) = \mathbf{E}_0 \delta(t), \quad (12)$$

integration of Eq. (10) gives, for $\mathbf{A}(t < 0) = 0$ under the sign conventions established above,

$$\mathbf{A}(t) = -c\mathbf{E}_0 \Theta(t). \quad (13)$$

The particular impulsive realization used in RMG is described separately in Sec. 3.

In the formal velocity-gauge representation, the periodic TDKS Hamiltonian for an elec-

tron is

$$\hat{H}_V(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL}}^V(t). \quad (14)$$

Here

$$\hat{V}_{\text{loc}}[\rho](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}[\rho](t) \quad (15)$$

contains all local terms, while the nonlocal pseudopotential is written separately. Since $q_e = -e$, the kinetic momentum in Eq. (14) is $\hat{\mathbf{p}} + e\mathbf{A}(t)/c$ for an electron. If a nonlocal pseudopotential is present, it must be transformed consistently with the same unitary gauge transformation,

$$\hat{V}_{\text{NL}}^V(t) = \hat{U}(t) \hat{V}_{\text{NL}} \hat{U}^\dagger(t), \quad \hat{U}(t) = \exp \left[\frac{iq_e}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right]. \quad (16)$$

The detailed consequences of Eq. (16) for the current operator are discussed in Sec. 2.2.

For a periodic system, the time-dependent Kohn–Sham orbitals are written in Bloch form,

$$\Psi_{n\mathbf{k}}(\mathbf{r}, t) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}, t), \quad (17)$$

where $u_{n\mathbf{k}}(\mathbf{r}, t)$ has the periodicity of the unit cell. Acting on a Bloch orbital, the canonical momentum becomes

$$\hat{\mathbf{p}} \Psi_{n\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{r}} (-i\hbar \nabla + \hbar \mathbf{k}) u_{n\mathbf{k}}. \quad (18)$$

The corresponding cell-periodic equation is

$$i\hbar \frac{\partial}{\partial t} u_{n\mathbf{k}}(\mathbf{r}, t) = \hat{H}_{\mathbf{k}}^V(t) u_{n\mathbf{k}}(\mathbf{r}, t), \quad (19)$$

with

$$\hat{H}_{\mathbf{k}}^V(t) = \frac{1}{2m_e} \left[-i\hbar \nabla + \hbar \mathbf{k} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL},\mathbf{k}}^V(t). \quad (20)$$

Equivalently, using $q_e = -e$, the kinetic operator contains $-i\hbar \nabla + \hbar \mathbf{k} + e\mathbf{A}(t)/c$. Equation (20) is the periodic counterpart of the finite-system length-gauge RT-TDDFT formulation used in our previous work,¹⁶ but with the position-dependent dipole perturbation replaced by a periodic velocity-gauge coupling.

Because the dipole moment is not a well-defined bulk observable under periodic boundary conditions, the natural time-dependent observable for periodic optical response is the macroscopic current rather than the molecular dipole moment. The construction of the current operator, including the nonlocal pseudopotential contribution, is given in Sec. 2.2; the extraction of optical response functions from the real-time current is described in Sec. 2.5.

2.2 Current Operator and Nonlocal Pseudopotentials

For a periodic system in the homogeneous-field limit, the optical response is naturally formulated in terms of the macroscopic charge-current density. We define

$$J_\alpha(t) = \frac{1}{V_{\text{cell}}} \sum_{\mathbf{k}, \sigma} w_{\mathbf{k}} \text{Tr} \left[\mathbf{P}_{\mathbf{k}}^\sigma(t) \hat{J}_{\mathbf{k}, \alpha}^\sigma(t) \right], \quad (21)$$

where V_{cell} is the volume of the periodic simulation cell, which may be a primitive cell or a supercell, $w_{\mathbf{k}}$ are the normalized Brillouin-zone weights satisfying $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$, $\mathbf{P}_{\mathbf{k}}^{\sigma}$ is the single-particle density matrix for spin channel σ , and

$$\hat{I}_{\alpha}(t) = q_e \hat{v}_{\alpha}(t) \quad (22)$$

is the charge-current operator. The trace in Eq. (21) is taken over the propagated single-particle subspace at fixed $\mathbf{k}$ and spin.

The velocity operator is most cleanly defined before introducing the cell-periodic Bloch representation. For the velocity-gauge Hamiltonian of Eq. (14),

$$\hat{v}_{\alpha}(t) = \frac{i}{\hbar} \left[\hat{H}_V(t), \hat{r}_{\alpha} \right]. \quad (23)$$

Since the local potential commutes with the position operator, the Hamiltonian containing a local potential and a nonlocal pseudopotential gives

$$\hat{v}_{\alpha}(t) = \frac{1}{m_e} \left[\hat{p}_{\alpha} - \frac{q_e}{c} A_{\alpha}(t) \right] + \frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_{\alpha} \right]. \quad (24)$$

Equivalently,

$$\frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_{\alpha} \right] = \frac{1}{i\hbar} \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL}}^V(t) \right], \quad (25)$$

which fixes the sign convention for the nonlocal contribution. Thus the formal charge-current operator is

$$\hat{I}_{\alpha}(t) = q_e \left\{ \frac{1}{m_e} \left[\hat{p}_{\alpha} - \frac{q_e}{c} A_{\alpha}(t) \right] + \frac{i}{\hbar} \left[\hat{V}_{\text{NL}}^V(t), \hat{r}_{\alpha} \right] \right\}. \quad (26)$$

The explicit $A_{\alpha}(t)$ dependence of the kinetic term gives the usual diamagnetic contribution,

$$\hat{I}_{\alpha}^{\text{dia}}(t) = -\frac{q_e^2}{m_e c} A_{\alpha}(t) \hat{1}, \quad (27)$$

while the nonlocal term is also A dependent through the gauge-transformed nonlocal pseudopotential.

The same nonlocal commutator appears when the velocity-gauge Hamiltonian is expanded to first order in the homogeneous vector potential. Using Eq. (16),

$$\hat{V}_{\text{NL}}^V(t) = \hat{U}(t) \hat{V}_{\text{NL}} \hat{U}^{\dagger}(t), \quad \hat{U}(t) = \exp \left[\frac{iq_e}{\hbar c} \mathbf{r} \cdot \mathbf{A}(t) \right], \quad (28)$$

one obtains

$$\hat{V}_{\text{NL}}^V(t) = \hat{V}_{\text{NL}} + \frac{iq_e}{\hbar c} \sum_{\alpha} A_{\alpha}(t) \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL}} \right] + \mathcal{O}(A^2). \quad (29)$$

The first-order nonlocal perturbation is therefore

$$\delta \hat{V}_{\text{NL}}^{(1)}(t) = -\frac{1}{c} \sum_{\alpha} A_{\alpha}(t) \hat{I}_{\alpha, \text{NL}}^{(0)}, \quad \hat{I}_{\alpha, \text{NL}}^{(0)} = \frac{q_e}{i\hbar} \left[\hat{r}_{\alpha}, \hat{V}_{\text{NL}} \right]. \quad (30)$$

Combining this term with the kinetic part of the minimal-coupling Hamiltonian gives the linear velocity-gauge perturbation

$$\delta\hat{H}^{(1)}(t) = -\frac{1}{c} \sum_{\alpha} A_{\alpha}(t) \hat{I}_{\alpha}^{(0)}, \quad (31)$$

where

$$\hat{I}_{\alpha}^{(0)} = q_e \left[\frac{\hat{p}_{\alpha}}{m_e} + \frac{i}{\hbar} [\hat{V}_{\text{NL}}, \hat{r}_{\alpha}] \right] = \frac{q_e}{m_e} \hat{p}_{\alpha} + \frac{q_e}{i\hbar} [\hat{r}_{\alpha}, \hat{V}_{\text{NL}}]. \quad (32)$$

Equations (26)–(32) show that the nonlocal-pseudopotential commutator is not only a correction to the measured current; it is also the nonlocal part of the linear velocity-gauge perturbation required for gauge consistency when nonlocal pseudopotentials are present.²⁵ More generally, a nonlocal one-particle operator present in the Hamiltonian contributes to the velocity through the corresponding commutator with the position operator.

For a separable norm-conserving pseudopotential,²⁶

$$\hat{V}_{\text{NL}} = \sum_{I,lm} |\beta_{I,lm}\rangle D_{I,lm} \langle\beta_{I,lm}|, \quad (33)$$

the coordinate-space matrix element of the commutator can be written as

$$\langle \mathbf{r} | [\hat{r}_{\alpha}, \hat{V}_{\text{NL}}] | \mathbf{r}' \rangle = (r_{\alpha} - r'_{\alpha}) V_{\text{NL}}(\mathbf{r}, \mathbf{r}'). \quad (34)$$

Equation (34) makes explicit that this term vanishes for a strictly local potential but is finite for a nonlocal pseudopotential.

The corresponding cell-periodic Bloch representation is obtained only after the full-space current operator has been defined. For $\Psi_{n\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}$, the canonical momentum becomes $-i\hbar\nabla + \hbar\mathbf{k}$ as in Eq. (18). The nonlocal operator at fixed $\mathbf{k}$ is

$$\hat{V}_{\text{NL},\mathbf{k}} = e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{V}_{\text{NL}} e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (35)$$

At zero external vector potential, the charge-current matrix in a cell-periodic basis is therefore

$$\begin{aligned} I_{ab,\mathbf{k},\alpha}^{(0),\sigma} &= \frac{q_e}{m_e} \langle u_{a\mathbf{k}}^{\sigma} | -i\hbar\nabla_{\alpha} + \hbar k_{\alpha} | u_{b\mathbf{k}}^{\sigma} \rangle_{\text{cell}} \\ &+ \frac{q_e}{i\hbar} \left\langle u_{a\mathbf{k}}^{\sigma} \left| [\hat{r}_{\alpha}, \hat{V}_{\text{NL},\mathbf{k}}^{\sigma}] \right| u_{b\mathbf{k}}^{\sigma} \right\rangle_{\text{cell}}. \end{aligned} \quad (36)$$

The first term contains both the gradient and Bloch $\hbar k_{\alpha}$ contributions, while the second term is the nonlocal-pseudopotential contribution. In the presence of a finite homogeneous vector potential, Eq. (26) gives the corresponding A -dependent current operator, including the diamagnetic kinetic term and the gauge-transformed nonlocal operator.

Equations (21) and (36) define the formal current observable used in the periodic RT-TDDFT theory. The implementation-specific construction of the current matrices, the particular impulsive realization of the velocity-gauge perturbation, and the conversion of raw current traces to optical response functions are described separately in Secs. 3 and 2.5.

2.3 Density-Matrix Time Propagation

The time propagation used in this work follows the density-matrix RT-TDDFT formulation developed for the finite-system implementation in our previous work,¹⁶ with the extension that the propagated matrices are now resolved by crystal momentum and collinear spin channel. For each $\mathbf{k}$ and spin σ , the propagation is carried out in a finite subspace $\mathcal{A}_{\mathbf{k}}^{\sigma}$ spanned by ground-state Kohn–Sham Bloch orbitals,

$$\Psi_{a\mathbf{k}}^{\sigma}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{a\mathbf{k}}^{\sigma}(\mathbf{r}), \quad a \in \mathcal{A}_{\mathbf{k}}^{\sigma}. \quad (37)$$

The one-particle density matrix in this basis is

$$P_{ab,\mathbf{k}}^{\sigma}(t) = \langle u_{a\mathbf{k}}^{\sigma} | \hat{\rho}^{\sigma}(t) | u_{b\mathbf{k}}^{\sigma} \rangle_{\text{cell}}. \quad (38)$$

The initial condition is the ground-state occupation matrix,

$$P_{ab,\mathbf{k}}^{\sigma}(0) = f_{a\mathbf{k}}^{\sigma} \delta_{ab}, \quad (39)$$

where $f_{a\mathbf{k}}^{\sigma}$ is the ground-state occupation of state a at $\mathbf{k}$ in spin channel σ .

The density matrix obeys the Liouville-von Neumann equation

$$i\hbar \frac{d\mathbf{P}_{\mathbf{k}}^{\sigma}(t)}{dt} = [\mathbf{H}_{\mathbf{k}}^{\sigma}(t), \mathbf{P}_{\mathbf{k}}^{\sigma}(t)], \quad (40)$$

where

$$H_{ab,\mathbf{k}}^{\sigma}(t) = \left\langle u_{a\mathbf{k}}^{\sigma} \left| \hat{H}_{\mathbf{k}}^{V,\sigma}(t) \right| u_{b\mathbf{k}}^{\sigma} \right\rangle_{\text{cell}} \quad (41)$$

is the matrix representation of the velocity-gauge Kohn–Sham Hamiltonian of Eq. (20). At nonzero $\mathbf{k}$, both $\mathbf{H}_{\mathbf{k}}^{\sigma}$ and $\mathbf{P}_{\mathbf{k}}^{\sigma}$ are generally complex Hermitian matrices.

Formally, Eq. (40) is solved by a unitary time-evolution operator,

$$\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) = \mathcal{U}_{\mathbf{k}}^{\sigma}(t + \Delta t, t) \mathbf{P}_{\mathbf{k}}^{\sigma}(t) \mathcal{U}_{\mathbf{k}}^{\sigma\dagger}(t + \Delta t, t), \quad (42)$$

with

$$\mathcal{U}_{\mathbf{k}}^{\sigma}(t + \Delta t, t) = \exp[\Omega_{\mathbf{k}}^{\sigma}(t + \Delta t, t)]. \quad (43)$$

The Magnus generator²⁷ is written as

$$\Omega_{\mathbf{k}}^{\sigma} = \Omega_{1,\mathbf{k}}^{\sigma} + \Omega_{2,\mathbf{k}}^{\sigma} + \cdots, \quad (44)$$

where

$$\Omega_{1,\mathbf{k}}^{\sigma} = -\frac{i}{\hbar} \int_t^{t+\Delta t} \mathbf{H}_{\mathbf{k}}^{\sigma}(\tau) d\tau, \quad (45)$$

and

$$\Omega_{2,\mathbf{k}}^{\sigma} = -\frac{1}{2\hbar^2} \int_t^{t+\Delta t} d\tau_1 \int_t^{\tau_1} d\tau_2 [\mathbf{H}_{\mathbf{k}}^{\sigma}(\tau_1), \mathbf{H}_{\mathbf{k}}^{\sigma}(\tau_2)]. \quad (46)$$

In the present method, the Magnus expansion is truncated after its first term, and the time integral of the Hamiltonian is evaluated using an endpoint-averaged, midpoint-equivalent

approximation,

$$\Omega_{\mathbf{k}}^{\sigma}(t + \Delta t, t) \simeq -\frac{i\Delta t}{2\hbar} [\mathbf{H}_{\mathbf{k}}^{\sigma}(t) + \mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)]. \quad (47)$$

This approximation has the time symmetry associated with midpoint or trapezoidal integration of the first Magnus term. The detailed construction of $\mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)$ in the self-consistent propagation loop is described in Sec. 3.

Because $\mathbf{H}_{\mathbf{k}}^{\sigma}$ is Hermitian, the generator $\Omega_{\mathbf{k}}^{\sigma}$ is anti-Hermitian and $\mathcal{U}_{\mathbf{k}}^{\sigma}$ is unitary. The action of the unitary transformation on the density matrix can therefore be written as a Baker–Campbell–Hausdorff nested-commutator series,

$$\begin{aligned} \mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) &= e^{\Omega_{\mathbf{k}}^{\sigma}} \mathbf{P}_{\mathbf{k}}^{\sigma}(t) e^{-\Omega_{\mathbf{k}}^{\sigma}} \\ &= \mathbf{P}_{\mathbf{k}}^{\sigma}(t) + [\Omega_{\mathbf{k}}^{\sigma}, \mathbf{P}_{\mathbf{k}}^{\sigma}(t)] + \frac{1}{2!} [\Omega_{\mathbf{k}}^{\sigma}, [\Omega_{\mathbf{k}}^{\sigma}, \mathbf{P}_{\mathbf{k}}^{\sigma}(t)]] + \dots \end{aligned} \quad (48)$$

Equivalently, defining $\mathbf{C}_{0,\mathbf{k}}^{\sigma} = \mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ and

$$\mathbf{C}_{n,\mathbf{k}}^{\sigma} = \frac{1}{n} [\Omega_{\mathbf{k}}^{\sigma}, \mathbf{C}_{n-1,\mathbf{k}}^{\sigma}], \quad (49)$$

one obtains

$$\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) = \sum_{n=0}^{\infty} \mathbf{C}_{n,\mathbf{k}}^{\sigma}. \quad (50)$$

This form applies equally to real Γ -point matrices and to the complex Hermitian matrices that arise for general $\mathbf{k}$ points. Exact unitary propagation conserves the trace, Hermiticity, and eigenvalues of each density matrix; therefore the single-particle occupations are preserved by the continuous evolution. In practical calculations these properties are preserved up to the numerical truncation and self-consistency errors of the propagation.

The propagated density matrices determine the real-space spin densities. When the propagated subspace is supplemented by frozen occupied states, the real-space spin density can be written as

$$\begin{aligned} \rho^{\sigma}(\mathbf{r}, t) &= \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{a,b \in \mathcal{A}_{\mathbf{k}}^{\sigma}} u_{a\mathbf{k}}^{\sigma}(\mathbf{r}) P_{ab,\mathbf{k}}^{\sigma}(t) u_{b\mathbf{k}}^{\sigma*}(\mathbf{r}) \\ &\quad + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{f \in \mathcal{F}_{\mathbf{k}}^{\sigma}} f_{f\mathbf{k}}^{\sigma} |u_{f\mathbf{k}}^{\sigma}(\mathbf{r})|^2. \end{aligned} \quad (51)$$

Here $\mathcal{F}_{\mathbf{k}}^{\sigma}$ denotes states that are not included in the propagated active subspace and remain fixed at their ground-state occupations. Equivalently, the same density can be written as a ground-state density plus the active-space density change,

$$\rho^{\sigma}(\mathbf{r}, t) = \rho_{\text{gs}}^{\sigma}(\mathbf{r}) + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{a,b \in \mathcal{A}_{\mathbf{k}}^{\sigma}} u_{a\mathbf{k}}^{\sigma}(\mathbf{r}) \Delta P_{ab,\mathbf{k}}^{\sigma}(t) u_{b\mathbf{k}}^{\sigma*}(\mathbf{r}), \quad (52)$$

where $\Delta \mathbf{P}_{\mathbf{k}}^{\sigma}(t) = \mathbf{P}_{\mathbf{k}}^{\sigma}(t) - \mathbf{P}_{\mathbf{k}}^{\sigma}(0)$. This form is especially useful for connecting the formal density reconstruction to an active-space implementation with frozen states.

The propagation is self-consistent because the Hamiltonian is a functional of the evolving density. The update cycle can be summarized as

$$\{\mathbf{P}_{\mathbf{k}}^{\sigma}(t)\} \rightarrow \{\rho^{\sigma}(\mathbf{r}, t)\} \rightarrow \hat{V}_{\text{H}}[\rho](t), \hat{V}_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}](t) \rightarrow \{\mathbf{H}_{\mathbf{k}}^{\sigma}(t)\} \rightarrow \{\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t)\}. \quad (53)$$

The two collinear spin channels are propagated as separate density matrices, but they are not independent: both contribute to the total density entering the Hartree potential, and the exchange-correlation potential in each spin channel depends on $\rho^{\uparrow}$ and $\rho^{\downarrow}$. The spin structure is discussed in more detail in Sec. 2.4.

2.4 Collinear Spin Extension

The spin-polarized extension used in the present work is collinear. A fixed spin quantization axis is chosen, and the spin index $\sigma \in \{\uparrow, \downarrow\}$ labels two spin channels. For each $\mathbf{k}$ point, the one-particle density matrix and Hamiltonian have a block-diagonal spin structure,

$$\mathbf{P}_{\mathbf{k}}(t) = \begin{pmatrix} \mathbf{P}_{\mathbf{k}}^{\uparrow}(t) & 0 \\ 0 & \mathbf{P}_{\mathbf{k}}^{\downarrow}(t) \end{pmatrix}, \quad \mathbf{H}_{\mathbf{k}}(t) = \begin{pmatrix} \mathbf{H}_{\mathbf{k}}^{\uparrow}(t) & 0 \\ 0 & \mathbf{H}_{\mathbf{k}}^{\downarrow}(t) \end{pmatrix}. \quad (54)$$

Thus there are no off-diagonal spinor blocks in the present formulation. The two spin channels are propagated as separate density matrices, each obeying the Liouville-von Neumann equation introduced in Sec. 2.3,

$$i\hbar \frac{d\mathbf{P}_{\mathbf{k}}^{\sigma}(t)}{dt} = [\mathbf{H}_{\mathbf{k}}^{\sigma}(t), \mathbf{P}_{\mathbf{k}}^{\sigma}(t)]. \quad (55)$$

The spin-resolved electron densities are reconstructed from $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ as described in Sec. 2.3. The total electron density is

$$\rho(\mathbf{r}, t) = \rho^{\uparrow}(\mathbf{r}, t) + \rho^{\downarrow}(\mathbf{r}, t). \quad (56)$$

In the collinear Kohn–Sham Hamiltonian, the local effective potential for spin channel σ can be written schematically as

$$\hat{V}_{\text{loc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}](t) = \hat{V}_{\text{ion}}^{\text{loc}} + \hat{V}_{\text{H}}[\rho](t) + \hat{V}_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}](t). \quad (57)$$

The Hartree potential depends on the total electron density $\rho = \rho^{\uparrow} + \rho^{\downarrow}$, while the exchange-correlation potential is spin-channel dependent and is a functional of both spin densities. In the absence of spin-orbit coupling, the nonlocal pseudopotential terms considered here are diagonal in the collinear spin basis and are treated as described in Secs. 2.1 and 2.2.

Equation (55) therefore represents separate propagation within each spin channel, but the channels are not physically independent. They are coupled self-consistently through the density-dependent potentials:

$$\{\mathbf{P}_{\mathbf{k}}^{\uparrow}, \mathbf{P}_{\mathbf{k}}^{\downarrow}\} \rightarrow \{\rho^{\uparrow}, \rho^{\downarrow}\} \rightarrow \hat{V}_{\text{H}}[\rho^{\uparrow} + \rho^{\downarrow}], \hat{V}_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}] \rightarrow \{\mathbf{H}_{\mathbf{k}}^{\uparrow}, \mathbf{H}_{\mathbf{k}}^{\downarrow}\}. \quad (58)$$

The same spin decomposition applies to the macroscopic charge current. For each spin

channel,

$$J_\alpha^\sigma(t) = \frac{1}{V_{\text{cell}}} \sum_{\mathbf{k}} w_{\mathbf{k}} \text{Tr} \left[\mathbf{P}_{\mathbf{k}}^\sigma(t) \hat{I}_{\mathbf{k},\alpha}^\sigma(t) \right], \quad (59)$$

where the charge-current operator is the spin-diagonal operator defined in Sec. 2.2. The total macroscopic charge-current density is the sum over the two collinear spin channels,

$$J_\alpha(t) = \sum_{\sigma=\uparrow,\downarrow} J_\alpha^\sigma(t). \quad (60)$$

The individual $J_\alpha^\sigma(t)$ may also be analyzed separately to obtain spin-resolved optical response within the fixed collinear basis.

The present formulation does not include off-diagonal spin coherences, transverse spin dynamics, general two-component spinors, noncollinear magnetization dynamics, or spin-orbit-induced spin mixing. Such extensions require a spinor Hamiltonian with off-diagonal spin blocks and lie outside the collinear spin-polarized RT-TDDFT framework used in this manuscript. This spin-resolved collinear formulation provides the theoretical basis for analyzing the spin-up and spin-down optical channels of the NV⁻ center discussed below.

2.5 Optical Response and Postprocessing

The velocity-gauge formulation developed above is not restricted to an impulsive perturbation. For a prescribed spatially homogeneous vector potential $\mathbf{A}(t)$, the formal Hamiltonian retains the external field for all times at which the chosen electromagnetic perturbation requires it,

$$\hat{H}_V(t) = \frac{1}{2m_e} \left[\hat{\mathbf{p}} - \frac{q_e}{c} \mathbf{A}(t) \right]^2 + \hat{V}_{\text{loc}}[\rho](t) + \hat{V}_{\text{NL}}^V(t). \quad (61)$$

Thus the same formal propagation can describe an impulsive excitation, a finite-duration pulse, or a persistent time-dependent driving field, provided the corresponding $\mathbf{A}(t)$ is used in the Hamiltonian and in the current operator of Sec. 2.2.

We use the Fourier-transform convention

$$f(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\omega) e^{i\omega t} d\omega. \quad (62)$$

With this convention,

$$\mathcal{F} \left[\frac{df}{dt} \right] = i\omega f(\omega), \quad (63)$$

assuming that boundary terms vanish or are treated separately. In the homogeneous velocity gauge,

$$\mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}, \quad \mathbf{A}(t) = \mathbf{A}(t_0) - c \int_{t_0}^t \mathbf{E}(t') dt', \quad (64)$$

and therefore

$$E_\alpha(\omega) = -\frac{i\omega}{c} A_\alpha(\omega). \quad (65)$$

A conventional broadband electric-field kick used in real-time linear response calcula-

tions^{16,28} is

$$E_\beta(t) = E_{0,\beta}\delta(t). \quad (66)$$

For $A_\beta(t < 0) = 0$, Eq. (64) gives

$$A_\beta(t) = -cE_{0,\beta}\Theta(t). \quad (67)$$

Thus, in the formally equivalent velocity-gauge representation, the vector potential remains at its post-kick value after the instantaneous electric-field impulse.

The time-dependent observable is the physical macroscopic charge-current density defined in Sec. 2.2. We write the induced current as

$$\Delta J_\alpha(t) = J_\alpha(t) - J_\alpha^{\text{eq}}, \quad (68)$$

where J_α^{eq} is the equilibrium current before the external perturbation. For the non-current-carrying equilibrium states considered here, the macroscopic equilibrium current vanishes in the exact ground state. Retaining the induced-current notation is nevertheless useful in numerical calculations, where finite $\mathbf{k}$ -point sampling and numerical errors may produce a small residual baseline.

In the linear-response regime,

$$\Delta J_\alpha(\omega) = \sum_\beta \sigma_{\alpha\beta}(\omega) E_\beta(\omega), \quad (69)$$

which defines the optical conductivity tensor $\sigma_{\alpha\beta}(\omega)$. Separate perturbations along linearly independent Cartesian directions may be used to construct the full tensor. For the electric-field impulse of Eq. (66), $E_\beta(\omega) = E_{0,\beta}$, and therefore

$$\sigma_{\alpha\beta}(\omega) = \frac{\Delta J_\alpha^{(\beta)}(\omega)}{E_{0,\beta}}, \quad (70)$$

where $\Delta J_\alpha^{(\beta)}$ denotes the current response to a kick polarized along direction β .

The dielectric response follows from the relation between current and macroscopic polarization. Defining the induced polarization by

$$\Delta J_\alpha(t) = \frac{d\Delta P_\alpha(t)}{dt}, \quad (71)$$

one obtains

$$\Delta J_\alpha(\omega) = i\omega \Delta P_\alpha(\omega), \quad \Delta P_\alpha(\omega) = \frac{\Delta J_\alpha(\omega)}{i\omega}. \quad (72)$$

Thus integration of the current in time corresponds formally to division by $i\omega$ in frequency space, with the zero-frequency component requiring separate treatment.

Using Gaussian electromagnetic units, the displacement field satisfies

$$D_\alpha(\omega) = E_\alpha(\omega) + 4\pi P_\alpha(\omega) = \sum_\beta \varepsilon_{\alpha\beta}(\omega) E_\beta(\omega). \quad (73)$$

Combining Eqs. (69) and (72) gives

$$\varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} - \frac{4\pi i}{\omega} \sigma_{\alpha\beta}(\omega). \quad (74)$$

With the Fourier convention of Eq. (62), the dissipative diagonal response is represented by $\text{Re } \sigma_{\alpha\alpha}(\omega)$ or, equivalently, by $-\text{Im } \varepsilon_{\alpha\alpha}(\omega)$. The sign of the imaginary part follows from the $e^{-i\omega t}$ transform convention used here.

The conversion from dielectric response to an absorption coefficient or absorbance depends on dimensionality and optical geometry. For a homogeneous bulk three-dimensional medium, one may introduce a complex refractive index $N_\alpha(\omega) = n_\alpha(\omega) + i\kappa_\alpha(\omega)$ along a principal axis, with $N_\alpha^2 = \varepsilon_{\alpha\alpha}$ and the sign of κ_α chosen to describe attenuation of the physical field. The corresponding bulk absorption coefficient is

$$\alpha_{\text{abs},\alpha}^{3\text{D}}(\omega) = \frac{2\omega}{c} \kappa_\alpha(\omega). \quad (75)$$

Absorbance additionally depends on the optical path length and sample geometry. Low-dimensional periodic systems, such as monolayers treated with supercell boundary conditions, require the appropriate geometry-dependent conversion. When only relative spectral features are compared, normalized absorption-like spectra should therefore be distinguished from quantitatively normalized conductivities, dielectric functions, absorption coefficients, or absorbances.

In a finite real-time calculation, the Fourier transform is evaluated over a finite propagation interval and is commonly combined with a window or damping function to control ringing and define an effective spectral broadening. A generic finite-time estimate of the induced current spectrum is

$$\Delta J_\alpha^T(\omega) = \int_0^T W(t) \Delta J_\alpha(t) e^{-i\omega t} dt, \quad (76)$$

where $W(t)$ is the chosen damping or window function. The finite propagation time and the choice of $W(t)$ determine the practical frequency resolution and broadening of the computed spectrum.

For collinear spin-polarized calculations, the total charge current is the sum of the spin-channel currents introduced in Sec. 2.4,

$$\Delta J_\alpha(t) = \sum_{\sigma=\uparrow,\downarrow} \Delta J_\alpha^\sigma(t). \quad (77)$$

The total optical conductivity follows from the total charge current in Eq. (69). Spin-channel-resolved spectra may also be formed from the separate $\Delta J_\alpha^\sigma(t)$ responses within the fixed collinear spin basis; this does not imply spin-flip transitions or noncollinear spin dynamics.

Equations (64)–(76) define the formal current-based optical response used in the present periodic velocity-gauge RT-TDDFT theory.²⁵ The numerical realization of the impulsive perturbation, the normalization of raw current traces, and the finite-time processing used in RMG are described in Sec. 3.

3 Implementation in RMG

The formal theory in Sec. 2.1–2.5 defines the periodic velocity-gauge RT-TDDFT formulation and the corresponding physical response functions. This section describes the current numerical realization in RMG. The implementation inherits the density-matrix propagation strategy of the previous finite-system RT-TDDFT module,¹⁶ and extends it to periodic boundary conditions, k-point sampling, collinear spin polarization, and current-based response analysis.

3.1 Numerical Implementation and Density-Matrix Propagation

A periodic RT-TDDFT calculation begins from a converged ground-state Kohn–Sham calculation. For each sampled $\mathbf{k}$ point and collinear spin channel σ , the ground-state cell-periodic Bloch orbitals are retained as a fixed basis for the real-time propagation. In this basis, RMG constructs and propagates finite active-space Hamiltonian and density matrices, $\mathbf{H}_{\mathbf{k}}^{\sigma}(t)$ and $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$, as introduced formally in Sec. 2.3. General $\mathbf{k}$ points lead to complex Hermitian matrices, while Γ -point calculations may use real-valued paths where applicable.

The implementation equations are evaluated internally in atomic units, with $\hbar = m_e = e = 1$ and $q_e = -1$. The speed of light is not set to unity by this convention. The relation between the explicit electromagnetic quantities in Sec. 2.1 and the numerical coefficient used by the VECTOR_POT mode is therefore treated separately in Sec. 3.2. The present periodic RT-TDDFT path is restricted to norm-conserving pseudopotentials.

The propagated orbital subspace is selected from the ground-state Kohn–Sham states. Its lower boundary is specified by the input parameter `tddft_start_state`; states below this boundary are not included in the propagated density matrix. Let $\{n_a\}$ denote the ground-state band indices retained in the active subspace and let N_{prop} be the number of propagated states. At the beginning of the real-time calculation,

$$H_{ab,\mathbf{k}}^{\sigma}(0) = \epsilon_{n_a\mathbf{k}}^{\sigma} \delta_{ab}, \quad P_{ab,\mathbf{k}}^{\sigma}(0) = f_{n_a\mathbf{k}}^{\sigma} \delta_{ab}, \quad (78)$$

where $\epsilon_{n_a\mathbf{k}}^{\sigma}$ and $f_{n_a\mathbf{k}}^{\sigma}$ are the ground-state eigenvalue and occupation of state n_a . Occupied states below the active lower boundary remain frozen at their ground-state density contribution. Occupied and unoccupied states inside the active subspace are included in $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ and can participate in the real-time response.

For a time step Δt , RMG uses the first term of the Magnus expansion with an endpoint-averaged Hamiltonian. In atomic units, the mathematically equivalent generator is

$$\Omega_{\mathbf{k}}^{\sigma}(t + \Delta t, t) \simeq -\frac{i\Delta t}{2} [\mathbf{H}_{\mathbf{k}}^{\sigma}(t) + \mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)]. \quad (79)$$

The code stores the Hermitian Hamiltonian integral separately from the factor of $-i$ used in the commutator propagation, but the resulting update is equivalent to

$$\mathbf{P}_{\mathbf{k}}^{\sigma}(t + \Delta t) = e^{\Omega_{\mathbf{k}}^{\sigma}} \mathbf{P}_{\mathbf{k}}^{\sigma}(t) e^{-\Omega_{\mathbf{k}}^{\sigma}}. \quad (80)$$

This unitary transformation is applied through the BCH nested-commutator expansion. The

complex periodic implementation uses this BCH propagation route.

The endpoint Hamiltonian $\mathbf{H}_{\mathbf{k}}^{\sigma}(t + \Delta t)$ is obtained self-consistently. RMG first predicts the endpoint Hamiltonian from the preceding Hamiltonian history. With this trial endpoint, Eq. (79) is used to propagate the density matrix, the real-space spin density is reconstructed, and the Hartree and spin-dependent exchange–correlation potentials are updated. The matrix elements of the resulting change in the local effective potential are then added to the endpoint Hamiltonian, and the cycle is repeated until the endpoint Hamiltonian is self consistent:

$$\mathbf{H}(t + \Delta t) \rightarrow \mathbf{\Omega} \rightarrow \mathbf{P}(t + \Delta t) \rightarrow \rho^{\sigma}(\mathbf{r}, t + \Delta t) \rightarrow V_{\text{H}}[\rho], V_{\text{xc}}^{\sigma}[\rho^{\uparrow}, \rho^{\downarrow}] \rightarrow \mathbf{H}(t + \Delta t).$$

The two collinear spin channels are propagated as separate density matrices, but the potentials are coupled through the total density entering the Hartree term and through the spin dependence of the exchange–correlation functional.

The real-space density reconstruction follows the active-space form of Sec. 2.3. RMG forms the active density-matrix change

$$\Delta P_{ab,\mathbf{k}}^{\sigma}(t) = P_{ab,\mathbf{k}}^{\sigma}(t) - f_{n_a\mathbf{k}}^{\sigma} \delta_{ab}, \quad (81)$$

and reconstructs

$$\rho^{\sigma}(\mathbf{r}, t) = \rho_{\text{gs}}^{\sigma}(\mathbf{r}) + \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{ab=1}^{N_{\text{prop}}} u_{n_a\mathbf{k}}^{\sigma}(\mathbf{r}) \Delta P_{ab,\mathbf{k}}^{\sigma}(t) u_{n_b\mathbf{k}}^{\sigma*}(\mathbf{r}). \quad (82)$$

The ordering of the orbital factors in Eq. (82) matches the density-matrix convention of Sec. 2.3: the ground-state density supplies the frozen occupied contribution and the reference occupation of the active subspace, while $\Delta \mathbf{P}_{\mathbf{k}}^{\sigma}(t)$ supplies the time-dependent active-space density change. This construction makes convergence with respect to the number of included virtual states an explicit numerical parameter of the RT-TDDFT calculation.

Exact unitary propagation preserves the trace, Hermiticity, and eigenvalues of each density matrix. In practical calculations these properties are maintained up to errors from the Magnus truncation, the finite BCH expansion, and the convergence tolerance of the endpoint self-consistency cycle.

3.2 External Perturbation and Current Evaluation

In the formal velocity-gauge theory, a prescribed homogeneous vector potential $\mathbf{A}(t)$ remains in the Hamiltonian for all times required by the external field. The present RMG `VECTOR_POT` mode implements a more specific discrete momentum-coupled excitation used to initiate the real-time response. We denote by κ_{α} the Cartesian numerical coefficient obtained from the `electric_field_tddft` input after the code’s coordinate conversion. This symbol denotes the coefficient actually used by the implementation; it is not identified here uniquely with a physical electric-field amplitude, vector-potential amplitude, A_{α}/c , or eA_{α}/c .

Before applying the perturbation, RMG constructs the kinetic/Bloch momentum-like

matrix

$$M_{ab,\mathbf{k},\alpha}^{\text{kin},\sigma} = \imath \langle (\nabla_\alpha + \imath k_\alpha) u_{n_a\mathbf{k}}^\sigma | u_{n_b\mathbf{k}}^\sigma \rangle_{\text{cell}}. \quad (83)$$

Using integration by parts under periodic boundary conditions, this is equivalent to the cell-periodic matrix element of $-\imath\nabla_\alpha + k_\alpha$ in atomic units. The explicit $\mathbf{k}$ term is the Bloch contribution to the momentum operator.

The implemented initial perturbation is then

$$\Delta \mathbf{H}_{\mathbf{k}}^{\text{code},\sigma} = \sum_{\alpha} \kappa_{\alpha} \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma}. \quad (84)$$

The verified construction order is

$$\text{VecPHmatrix}() \rightarrow \text{VECTOR_POT perturbation} \rightarrow \text{CurrentNlpp}().$$

Consequently, the perturbing operator in Eq. (84) contains the kinetic/Bloch contribution only. The nonlocal-pseudopotential commutator correction does not enter the initial perturbing kick in the present implementation.

The perturbing contribution enters the first accepted midpoint propagator through the initial Hamiltonian endpoint and is absent from the converged self-consistent Hamiltonian without the external perturbing term at subsequent times. Since the first Magnus generator uses an endpoint-averaged Hamiltonian, the effective integrated first-order numerical action of the perturbation is

$$\int dt \Delta \mathbf{H}_{\mathbf{k}}^{\text{code},\sigma}(t) \approx \frac{\Delta t}{2} \sum_{\alpha} \kappa_{\alpha} \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma}. \quad (85)$$

Equation (85) describes the current discrete numerical realization. It should not be read as a statement that the code applies a mathematically exact Dirac-delta vector potential or the formally persistent step-like vector potential of Sec. 2.5.

After the perturbation is applied, RMG adds the nonlocal pseudopotential contribution to the matrices used for current evaluation. The routine `CurrentNlpp()` calls the same directional nonlocal-operator path used for the projector commutator. Expressed in the sign convention of Sec. 2.2, the atomic-unit nonlocal velocity contribution is $\imath[\hat{V}_{\text{NL}}, \hat{r}_\alpha]$, or equivalently $(1/\imath)[\hat{r}_\alpha, \hat{V}_{\text{NL}}]$. The corresponding raw matrix contribution is written as

$$M_{ab,\mathbf{k},\alpha}^{\text{NL},\sigma} = \frac{1}{\imath} \langle u_{n_a\mathbf{k}}^\sigma | [\hat{r}_\alpha, \hat{V}_{\text{NL},\mathbf{k}}^\sigma] | u_{n_b\mathbf{k}}^\sigma \rangle_{\text{cell}}. \quad (86)$$

Equivalently, this term may be written as $\imath \langle u_{n_a\mathbf{k}}^\sigma | [\hat{V}_{\text{NL},\mathbf{k}}^\sigma, \hat{r}_\alpha] | u_{n_b\mathbf{k}}^\sigma \rangle_{\text{cell}}$. This is the same sign convention used in the formal current matrix of Eq. (36), excluding the charge factor q_e and the volume normalization that are not included in the raw RMG matrix.

Thus the raw current-evaluation matrix used by RMG is

$$\mathbf{M}_{\mathbf{k},\alpha}^{\text{raw},\sigma} = \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma} + \mathbf{M}_{\mathbf{k},\alpha}^{\text{NL},\sigma}. \quad (87)$$

For calculations using the simple DFT+U implementation, the same directional operator-

construction path also includes the corresponding Hubbard-projector contribution,

$$\mathbf{M}_{\mathbf{k},\alpha}^{\text{raw},\sigma} = \mathbf{M}_{\mathbf{k},\alpha}^{\text{kin},\sigma} + \mathbf{M}_{\mathbf{k},\alpha}^{\text{NL},\sigma} + \mathbf{M}_{\mathbf{k},\alpha}^{U,\sigma}. \quad (88)$$

This statement identifies the implemented code path; it does not by itself establish that the DFT+U contribution has been independently proven to be the full gauge-consistent commutator for a Hubbard functional.

The distinction between perturbation and measurement is therefore important. The current RMG `VECTOR_POT` perturbation uses the kinetic/Bloch momentum-like matrix in Eq. (84), whereas the matrix subsequently used to evaluate the current-like response contains the nonlocal pseudopotential correction and, where active, the simple DFT+U projector contribution. The implementation also does not explicitly construct the A -dependent diamagnetic current operator appearing in the formal charge-current operator of Sec. 2.2.

3.3 Periodic, Spin-Resolved, and Parallel Implementation

The periodic implementation propagates a separate active-space Hamiltonian and density matrix for each local $\mathbf{k}$ point and collinear spin channel. Thus $\mathbf{H}_{\mathbf{k}}^{\sigma}(t)$, $\mathbf{P}_{\mathbf{k}}^{\sigma}(t)$, and the corresponding momentum/current-like matrices are stored and updated independently for each $\mathbf{k}$ handled by a given MPI task group. General $\mathbf{k}$ points require complex Bloch orbitals and complex Hermitian matrices, while Γ -point calculations may use real-valued orbital and matrix paths where applicable. The Bloch $\mathbf{k}$ contribution to the kinetic momentum matrix is included in the operator construction described in Sec. 3.2.

The density and observable reductions use the Brillouin-zone weights associated with the sampled $\mathbf{k}$ points. During the self-consistent endpoint update, RMG reconstructs a density contribution from each local $\mathbf{k}$ point, multiplies it by the corresponding weight, and sums over the distributed $\mathbf{k}$ set. The same weighted reduction is used for current-like vector observables. After the $\mathbf{k}$ -point and matrix-distribution reductions, the vector current is passed through the RMG symmetry machinery for vector symmetrization.

For collinear spin-polarized calculations, RMG propagates the two spin channels as separate matrix problems. Each spin channel has its own $\mathbf{P}_{\mathbf{k}}^{\uparrow}$ or $\mathbf{P}_{\mathbf{k}}^{\downarrow}$ and its own spin-dependent Hamiltonian. The implementation does not introduce off-diagonal spinor blocks, spin flips, transverse spin dynamics, or spin-orbit induced spin mixing in the collinear RT-TDDFT path used here. The channels are nevertheless coupled self-consistently: after the $\mathbf{k}$ -weighted density reconstruction, the spin density available on a given spin partition is combined with the opposite-spin density before the Hartree and exchange–correlation potentials are evaluated. The Hartree update depends on the total density, while the exchange–correlation update uses the spin-dependent functional form with both spin densities.

Current-like observables are accumulated separately for each spin channel in the implementation. In the notation of Secs. 2.4 and 2.5, the physical total charge-current response is obtained from the spin-channel contributions,

$$\Delta J_{\alpha}(t) = \Delta J_{\alpha}^{\uparrow}(t) + \Delta J_{\alpha}^{\downarrow}(t). \quad (89)$$

The separate spin-channel traces can also be retained for spin-resolved analysis within the

fixed collinear basis. This organization is used later to analyze the spin-up and spin-down optical channels of the NV^- center.

The parallel decomposition combines the real-space domain decomposition inherited from ground-state RMG with additional parallelism over $\mathbf{k}$ points, spin channels, and dense active-space matrices. Real-space operations such as density reconstruction, application of local potentials, and grid integrations are distributed over the real-space grid. The active-space Hamiltonian and density matrices are distributed using the matrix backend selected for the run, including ScaLAPACK and tiled matrix-multiplication paths. The refactored `rmg::tddft<OrbitalType,MatrixType>` class provides CPU and GPU execution paths through templated orbital and matrix types.

This organization exposes independent work over $\mathbf{k}$ points and spin channels while retaining the locality of the real-space RMG operations and the efficiency of distributed dense matrix algebra. It is this combination of real-space decomposition, $\mathbf{k}$ /spin decomposition, matrix parallelism, and accelerator support that makes the periodic density-matrix RT-TDDFT implementation applicable to large supercells, including the 4095-atom NV^- -center calculations discussed below.

3.4 Optical Spectra and Implementation-Specific Processing

The formal response relations in Sec. 2.5 define the optical conductivity and dielectric tensor in terms of the physical macroscopic charge-current density. RMG writes spin-resolved raw current-like traces, for which the available current-postprocessing workflow produces optical spectra in arbitrary units.

Using the raw current-evaluation matrix defined in Sec. 3.2, the quantity written by RMG for spin channel σ can be represented schematically as

$$R_\alpha^\sigma(t_n) = \mathcal{S}_\alpha \left\{ \sum_{\mathbf{k}} w_{\mathbf{k}} \text{Re Tr} [\mathbf{P}_{\mathbf{k}}^\sigma(t_n) \mathbf{M}_{\mathbf{k},\alpha}^{\text{raw},\sigma}] \right\}, \quad (90)$$

where $\mathcal{S}$ denotes the vector symmetrization applied after the parallel reductions. The raw trace includes $\mathbf{k}$ -point weights and the reduction over distributed $\mathbf{k}$ -point and matrix partitions. It is written separately for each spin channel and does not explicitly include multiplication by q_e , division by V_{cell} , or the sum over spin channels.

Consequently, $R_\alpha^\sigma(t)$ should not be identified directly with the physical macroscopic charge-current density $J_\alpha^\sigma(t)$ of Sec. 2.2. If the raw matrix is interpreted as a zero-field velocity-like matrix in atomic units, conversion to the formal current density would require, at minimum, the electron-charge factor, the cell-volume normalization, the spin summation for the total current, and the corresponding unit conversion:

$$J_\alpha(t) \sim \frac{q_e}{V_{\text{cell}}} \sum_{\sigma=\uparrow,\downarrow} R_\alpha^\sigma(t), \quad (91)$$

with $q_e = -1$ in the internal atomic-unit convention. This is a schematic relation identifying missing normalization factors, not an operation currently applied by the RMG current-output routine. The code also prints a ground-state current as a diagnostic header entry,

but this value is not directly subtracted from the written time trace. In a fresh calculation, the first numerical current row is written after the first accepted propagation step, using the step-counter time label.

For current-based spectral analysis, the available RMG workflow first removes a numerical baseline from the sampled trace. For current files, it then applies a cumulative summation of the current-like signal, which acts as an implementation-specific discrete current-to-polarization integration. The resulting time series may be detrended and damped or windowed, Fourier transformed, normalized by the raw perturbation coefficient, and smoothed to produce the plotted spectrum. This processing is useful for comparing spectral shapes and peak positions, but the available workflow does not preserve the full signed complex phase of the response. It therefore does not by itself produce a quantitatively normalized complex conductivity or dielectric function.

The spectra presented below are reported in arbitrary units and are interpreted primarily through relative peak positions, spectral shape, polarization dependence, spin dependence, and active-space convergence rather than absolute optical intensities. A fully normalized conductivity or dielectric response can be obtained from the formal relations of Sec. 2.5, but doing so requires retaining the complex Fourier response and applying the physical charge, volume, spin-summation, field-normalization, and unit-conversion factors explicitly.

4 Validation of Periodic and Spin-Polarized RT-TDDFT

We first validate the periodic and spin-polarized RT-TDDFT implementation in RMG by comparing current-derived optical spectra with independent real-time calculations performed with `Octopus`, an established real-space RT-TDDFT code for finite and extended systems.²⁹ The benchmark set was chosen as a progression of increasing complexity: a one-dimensional periodic hydrogen-dimer reference, pristine monolayer hBN, bulk Si, and spin-polarized CrI₃. The purpose of these comparisons is not to establish quantitatively normalized optical conductivities or dielectric functions, but to test whether RMG reproduces the principal spectral features obtained from an independent RT-TDDFT implementation at a comparable semilocal-DFT level. Accordingly, Fig. 1 reports optical spectra in arbitrary units and the comparison emphasizes peak positions, dominant spectral structures, overall spectral envelopes, and polarization or spin dependence where applicable.

The one-dimensional hydrogen-dimer system provides the simplest periodic reference. The model consists of hydrogen dimers repeated along one lattice direction with large transverse vacuum, and therefore tests the basic k-resolved propagation and current-response machinery without the additional complications of a multidimensional crystal. RMG reproduces the sequence of major peaks obtained from the `Octopus` calculation over the plotted energy range, providing a basic cross-code validation of the periodic propagation and current-response implementation.

Pristine monolayer hBN provides a genuinely two-dimensional periodic benchmark. The RMG calculation uses a non-spin-polarized monolayer slab with vacuum normal to the layer and dense in-plane k-point sampling. Similar monolayer hBN tests have been used in recent periodic velocity-gauge RT-TDDFT implementation work.³⁰ The RMG and `Octopus` spectra show a similar low-energy onset and similar dominant structures in the main spec-

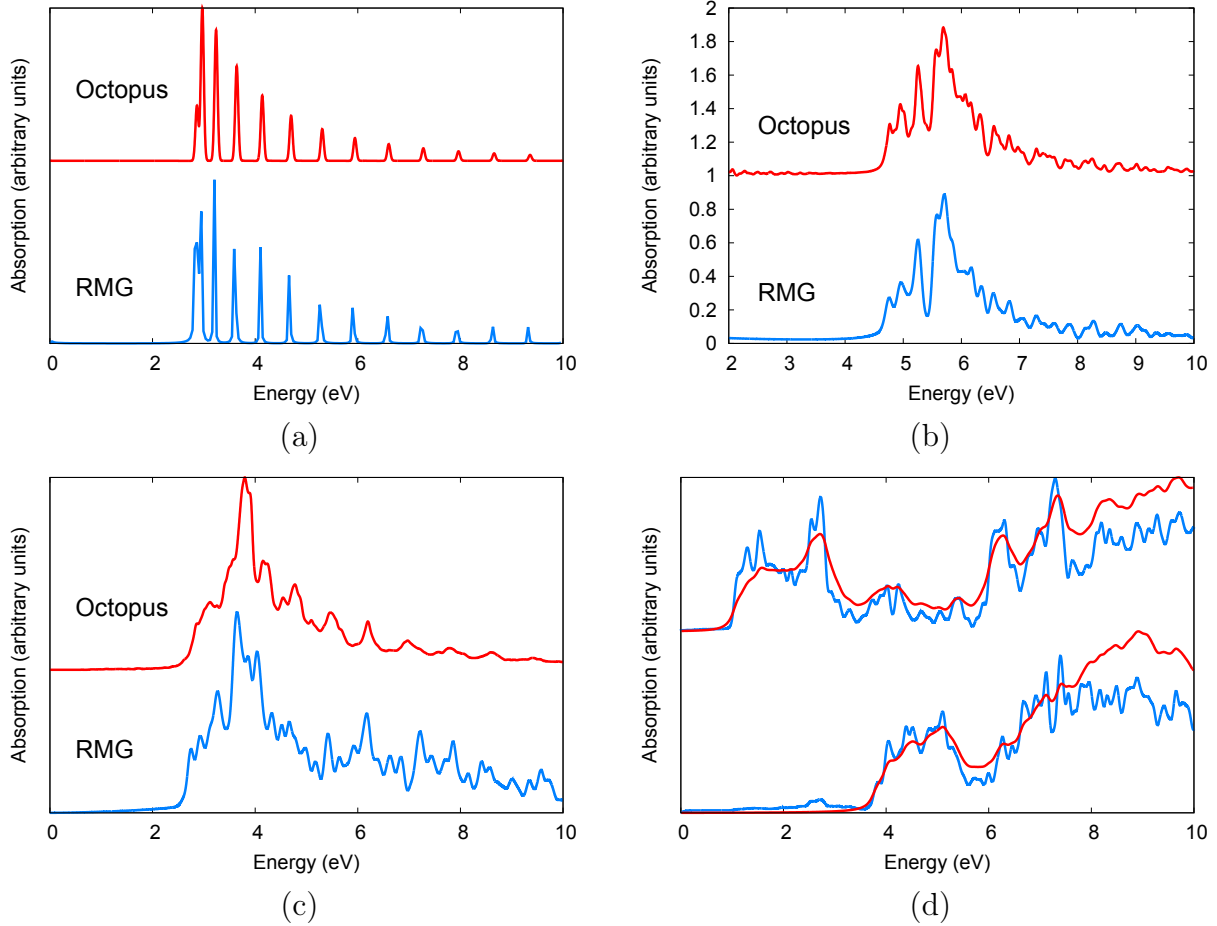


Figure 1: Optical spectra in arbitrary units from periodic RT-TDDFT calculations with RMG and independent `Octopus` reference calculations for systems of increasing complexity: (a) a one-dimensional periodic hydrogen-dimer reference, (b) pristine monolayer hBN, (c) bulk Si, and (d) spin-polarized CrI_3 . The comparison emphasizes peak positions and dominant spectral features rather than absolute intensity.

tral envelope, validating the two-dimensional periodic implementation and the in-plane current response for an insulating material. As expected for semilocal PBE-level calculations, this benchmark should not be interpreted as a quantitative description of the strongly excitonic experimental optical spectrum of low-dimensional BN materials, where long-range electron-hole interactions are known to be important.³¹ Detailed spin-polarized RT-TDDFT applications to hBN color centers are treated separately in a companion study.

Bulk Si then tests the full three-dimensional periodic implementation, including dense k-point sampling and the complex matrix path required away from Γ . The spectra from RMG and `Octopus` show similar low-energy behavior and reproduce the dominant structure in the approximately 3–4 eV region, while somewhat larger differences are visible in the higher-energy fine structure. This level of agreement is consistent with the role of Si as a standard benchmark for periodic velocity-gauge RT-TDDFT implementations.³²

Finally, CrI_3 provides the most demanding member of the general benchmark set: a spin-polarized periodic calculation including DFT+ U for the Cr 3d states. The available RMG input is collinear spin-polarized and includes a $U = 3.0$ eV correction for the Cr 3d states.

The comparison in Fig. 1 shows that RMG captures the main spectral structures obtained in the independent calculation for this magnetic material. We interpret this benchmark as a test of the spin-polarized periodic RT-TDDFT workflow, not as a complete treatment of spin-orbit-coupled or excitonic magneto-optical physics in CrI_3 , which requires physics beyond the present collinear PBE+U-level workflow.³³

Taken together, these four benchmarks show that the RMG implementation reproduces the principal spectral features of an independent RT-TDDFT code across one-dimensional, two-dimensional, three-dimensional, and spin-polarized periodic systems. The following NV^- -center calculations provide an additional physics-specific validation layer by testing defect orbital character, polarization selection rules, spin-channel selectivity, active-space convergence, and supercell-size effects in large diamond supercells.

4.1 JACEK: Large-Scale Demonstration: NV Center in Diamond

As a final demonstration, we apply the periodic spin-polarized RT-TDDFT implementation to the negatively charged nitrogen-vacancy (NV^-) center in diamond, one of the most studied point defects in solid-state physics.^{34–36} The ground state of NV^- center is an optically addressable spin-triplet located within the diamond band gap^{35–37} Because of its optical addressability and long coherence times at room temperature the NV^- center is an attractive system for several emerging quantum technologies, including single-photon emitter, nanoscale magnetometry, quantum sensing, quantum information processing.^{38–43}

This system tests several capabilities simultaneously: spin-polarized real-time propagation, optical response of a localized defect in a three-dimensional periodic host, convergence toward the dilute-defect limit, as well as computational scaling to thousands of atoms. The purpose of this section is not to provide a complete quantitative description of NV^- photophysics. Instead, the present calculations examine the electronic optical response of defect in the energy region of the experimental zero-phonon line (ZPL) and use the NV^- center to demonstrate how polarization selection rules, periodic defect-image interactions, and active-space truncation affect large-scale periodic RT-TDDFT calculations.

4.1.1 Electronic structure and simulation setup for NV^-

The NV^- center consists of a substitutional nitrogen atom adjacent to a carbon vacancy, with one additional electron trapped at the defect (Figure 2a). The defect axis is oriented along the crystallographic [111] direction, which defines the C_3 symmetry axis of the approximately C_{3v} local defect environment. The defect states are located within the diamond band gap. The lower a_1 -like defect orbital has substantial nitrogen character together with contributions from the three carbon atoms adjacent to the vacancy, whereas the nearly degenerate e manifold is primarily localized on these three carbon atoms, as shown in Figure 2c–e. Because pristine diamond has a wide band gap of about 5.5 eV and is transparent throughout the visible range, optical absorption and emission below this energy arise from defect states; for NV^- , the $^3A_2 \rightarrow ^3E$ transition has an experimental zero-phonon line at 1.945 eV, corresponding to 637 nm red light.

All NV^- calculations were performed here with the RMG code using periodic boundary conditions, the PBE exchange-correlation functional, and the NC_accuracy norm-conserving

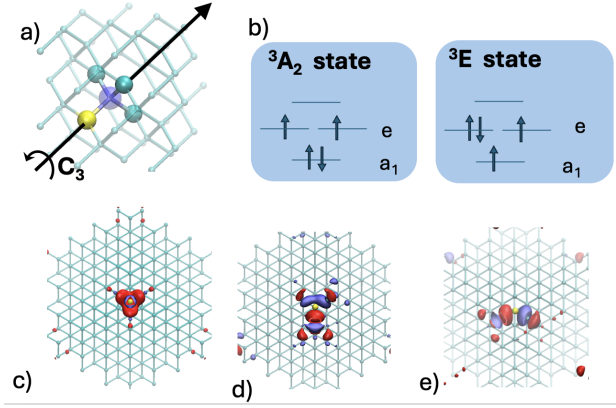


Figure 2: Electronic structure of the negatively charged nitrogen-vacancy (NV^-) center in diamond. (a) Local atomic structure of the defect, consisting of a substitutional nitrogen atom (green) adjacent to a carbon vacancy (blue transparent). The black arrow indicates the $[111]$ direction, which defines the C_3 symmetry axis of the approximately C_{3v} defect environment. (b) Schematic occupation of the defect levels in the triplet ground state, 3A_2 (configuration $a_1^2 e_1^1 e_2^1$), and in the optically excited triplet 3E state (configuration $a_1^1 e_1^2 e_2^1$). The defect-associated optical transition in the experimental zero-phonon-line energy region is dominated by promotion of a minority-spin electron from the occupied a_1 -like defect orbital into the e manifold. (c–e) Representative real-space isosurfaces of Kohn–Sham defect orbitals localized in the diamond band gap (view along the C_3 axis: (c) the occupied a_1 -like orbital and (d,e) the two components of the nearly degenerate e manifold. Red and blue isosurfaces denote opposite phases of the Kohn–Sham orbitals.

pseudopotentials. The diamond lattice constant was first optimized for bulk diamond using the same functional and pseudopotentials, giving $a = 3.5725125 \text{ \AA}$. This optimized lattice constant was then kept fixed for all NV^- defect calculations.

Three single-defect supercell sizes were considered, denoted NV- $x2$ (small), NV- $x4$ (medium), and NV- $x8$ (large). These systems were constructed from $2 \times 2 \times 2$, $4 \times 4 \times 4$, and $8 \times 8 \times 8$ repetitions of the conventional face-centered cubic (fcc) diamond cell, corresponding to 64, 512, and 4096 diamond lattice sites before formation of the defect. Previous supercell studies of NV^- in diamond have shown that relatively large cells, on the order of the 512-site NV- $x4$ model, are needed to reduce spurious interactions between periodic defect images.^{35–37} Here we directly test this finite-size behavior by extending the real-time calculations to the 4096-site NV- $x8$ cell. The same 4096-site host also enables explicit simulations of periodic arrays of NV^- centers, including collinear parallel and antiparallel defect-spin arrangements, which are used below as an additional demonstration of spin-resolved periodic RT-TDDFT. The actual defect supercells contained 63, 511, and 4095 atoms, respectively. The negatively charged supercells were treated using a uniform compensating background charge to maintain overall charge neutrality under periodic boundary conditions.

The real-space grid was scaled with the supercell size so that all three calculations used the same grid spacing. The NV- $x2$, NV- $x4$, and NV- $x8$ systems used 48^3 , 96^3 , and 192^3 grid points, respectively, corresponding to a grid spacing of $0.281 a_0$. This grid corresponds to equivalent wavefunction and charge-density cutoffs of 124.7 and 498.9 Ry, respectively; the charge-density grid is two times finer in each direction.

After construction of the charged spin-polarized defect cells, all atomic coordinates were relaxed at fixed lattice vectors. For each supercell size, the spin-polarized triplet solution was

found to be lower in energy than the corresponding singlet solution, and the relaxed triplet geometries were used as the starting point for the RT-TDDFT simulations. All real-time calculations used spin-collinear density-matrix propagation with a time step of 1.0 atomic unit of time and production trajectories of 40000 time steps, corresponding to approximately 1.0 ps.

Table 1: Single-defect NV⁻ supercells used in the RT-TDDFT calculations. The number of diamond sites refers to the pristine diamond supercell before creating the NV center. The actual number of atoms is one smaller after replacing one C atom by N and removing a nearest-neighbor C atom to form the vacancy.

System	Supercell	Sites	Atoms	Grid	h (a_0)	k -point sampling
NV- $x2$ (small)	$2 \times 2 \times 2$	64	63	48^3	0.281	$\Gamma, 4 \times 4 \times 4$
NV- $x4$ (medium)	$4 \times 4 \times 4$	512	511	96^3	0.281	$\Gamma, 2 \times 2 \times 2$
NV- $x8$ (large)	$8 \times 8 \times 8$	4096	4095	192^3	0.281	Γ

4.1.2 Polarization selection rules and spin selectivity

The polarization-resolved RT-TDDFT spectra provide a direct test of the local C_{3v} selection rules of the NV⁻ center. This analysis was performed for the medium-size single-defect NV- $x4$ supercell (511 atoms) at the Γ point. The TDDFT active space consisted of 1024 occupied and 3200 virtual orbitals (approximately 6 per each C atom), overall total 4324 orbitals. Three impulsive electric-field polarizations were considered: one parallel to the NV axis,

$$\hat{\mathbf{e}}_{\parallel} = \frac{1}{\sqrt{3}}(1, 1, 1), \quad (92)$$

and two mutually orthogonal directions perpendicular to the NV axis,

$$\hat{\mathbf{e}}_{\perp,1} = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \hat{\mathbf{e}}_{\perp,2} = \frac{1}{\sqrt{6}}(1, 1, -2). \quad (93)$$

The calculated spectra show that the defect-associated peak in the experimental zero-phonon-line energy region is absent for the polarization parallel to the [111] defect axis, while it appears clearly for both transverse polarizations, $\hat{\mathbf{e}}_{\perp,1}$ and $\hat{\mathbf{e}}_{\perp,2}$. See Figure 3.

This behavior follows directly from the symmetry of the ${}^3A_2 \rightarrow {}^3E$ transition. In the local C_{3v} point group, the velocity (or dipole operator for length gauge case) component parallel to the defect axis transforms as A_1 , whereas the two transverse velocity (or dipole operator for length gauge case) components transform together as the two-dimensional E representation. The electric-dipole matrix element is symmetry allowed only when the direct product of the initial state, dipole operator, and final state contains the totally symmetric representation A_1 . For axial polarization,

$$A_2 \otimes A_1 \otimes E = E, \quad (94)$$

which does not contain A_1 , and the transition is forbidden. For transverse polarization,

$$A_2 \otimes E \otimes E = A_1 \oplus A_2 \oplus E, \quad (95)$$

which contains A_1 , and the transition is allowed. The computed polarization dependence is therefore consistent with the expected selection rule that the optical transition dipole for the $\text{NV}^- \ ^3A_2 \rightarrow \ ^3E$ transition lies perpendicular to the NV axis.

The same spectra also demonstrate spin selectivity of the defect-region optical response. In the collinear spin convention used in the present calculations, the peak in the zero-phonon-line energy region appears predominantly in the spin-down channel. Equivalently, this is the minority-spin channel of the triplet configuration shown in Fig. 2b. The dominant excitation promotes a minority-spin electron from the occupied a_1 -like defect orbital into the e manifold, producing the configuration schematically shown for the optically excited triplet state. Because the electric-dipole perturbation does not flip spin, the transition appears in the spin channel associated with this minority-spin $a_1 \rightarrow e$ excitation. This spin-resolved selection rule provides a stringent test of the spin-collinear RT-TDDFT implementation for localized defect states in a periodic host.

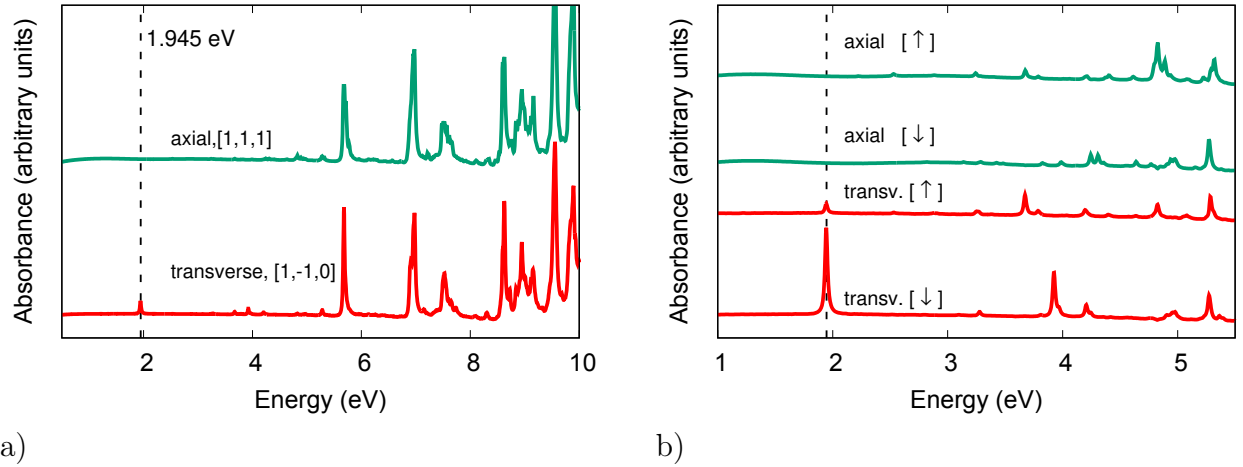


Figure 3: Polarization and spin selectivity of the NV^- defect optical response from spin-collinear RT-TDDFT for the 512-site $\text{NV-}x4$ supercell. (a) Spin-summed spectra for axial and transverse light polarizations. The axial spectrum corresponds to a kick polarized parallel to the NV axis, $\mathbf{E} \parallel [111]$, while the transverse spectrum corresponds to $\mathbf{E} \parallel [1\bar{1}0]$, perpendicular to the NV axis. The dashed vertical line marks the experimental NV^- zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The defect-associated peak near this energy is strongly enhanced for transverse polarization and suppressed for axial polarization, consistent with the C_{3v} selection rule for the $\ ^3A_2 \rightarrow \ ^3E$ transition. (b) Expanded view of the sub-gap optical response below approximately 5.5 eV, corresponding to the transparency window of pristine diamond. In this energy range, optical features arise from the NV^- defect states rather than from intrinsic band-to-band absorption of the diamond host. The spin-resolved spectra show that the transverse response near the ZPL energy is dominated by the spin-down, minority-spin channel, consistent with the dominant $a_1 \rightarrow e$ defect excitation shown schematically in Fig. 2. The spectra are vertically shifted for clarity.

4.1.3 Active-space convergence of the defect excitation

The selection-rule analysis above used the medium-size single-defect NV- $x4$ supercell at the Γ point with the TDDFT active space including full occupied orbitals manifold and 3200 virtual orbitals. Here we use the same system and transverse polarization to examine how the defect-associated peak depends on the active propagation manifold used in the RT-TDDFT calculation. As the system size increases, the number of occupied and unoccupied Kohn–Sham states included in the density-matrix propagation can grow rapidly, making the computational cost prohibitive. Similarly, one might expect the low-energy defect excitation to be only weakly sensitive to deep occupied states or to high-energy virtual states, since the relevant NV[−] defect states are localized in the diamond band gap. We investigate the effect of the TDDFT active space construction on position of the experimental zero phonon line 1.945 eV peak to quantify the effects of this effect numerically in terms of fraction of the occupied orbitals, $f_{\text{occ}} = \frac{N_{\text{occ}}^{\text{act}}}{N_{\text{occ}}^{\text{full}}}$, and the number of virtual states per diamond site, $n_{\text{virt}} = \frac{N_{\text{virt}}}{N_{\text{site}}}$, used to construct the TDDFT active space. Here, $N_{\text{occ}}^{\text{full}}$ represents the total number of occupied orbitals, $N_{\text{occ}}^{\text{act}}$ is the number of occupied states retained in the propagation, N_{virt} is the number of unoccupied states included in the active manifold and N_{site} is the total number of diamond lattice sites before formation of the defect.

For the 512-site NV- $x4$ supercell used in this convergence test, $N_{\text{occ}}^{\text{full}} = 1024$ and $N_{\text{site}} = 512$. The occupied-space tests removed the lowest 0, 200, 400, 512, 600, and 800 occupied Kohn–Sham states from the active propagation manifold. This corresponds to retaining 1024, 824, 624, 512, 424, and 224 occupied states, or approximately 100%, 80%, 60%, 50%, 40%, and 20% of the occupied manifold, respectively. The virtual-space tests used $N_{\text{virt}} = 512$, 1024, 2048, and 3200, corresponding to $n_{\text{virt}} = 1, 2, 4$, and 6.25 virtual states per diamond site.

Figure 4 shows the effect of occupied- and virtual-space truncation on the position of the defect-associated peak near the experimental ZPL energy. Reducing f_{occ} shifts the peak to higher energy, showing that the deep occupied manifold cannot be discarded without changing the computed defect excitation. Conversely, increasing n_{virt} systematically shifts the peak to lower energy, with the peak position approaching a saturated value as the virtual space is enlarged. Physically, this behavior indicates that the localized defect excitation is dressed by the polarizable diamond environment and by the self-consistent time-dependent response of the electron density. Truncating either the occupied or unoccupied part of the propagation manifold suppresses electronic relaxation and screening around the particle-hole excitation, producing a measurable blueshift

Furthermore, the two convergence tests show different sensitivities. At fixed $n_{\text{virt}} = 6.25$, reducing the retained occupied fraction from 100% to about 50% produces only a modest blue shift of the defect-associated peak, while retaining only about 20% of the occupied space leads to a larger shift. By contrast, at fixed $f_{\text{occ}} = 100\%$, increasing the virtual active space from $n_{\text{virt}} = 1$ to 4–6.25 virtual states per site produces a substantially larger red shift and brings the peak close to the experimental ZPL energy. Thus, for this NV- $x4$ test case, the unoccupied portion of the active space is the primary convergence bottleneck for the peak position, while the occupied manifold can be moderately truncated with a smaller, but still measurable, effect.

In summary, the peak position is more sensitive to the size of the virtual active space

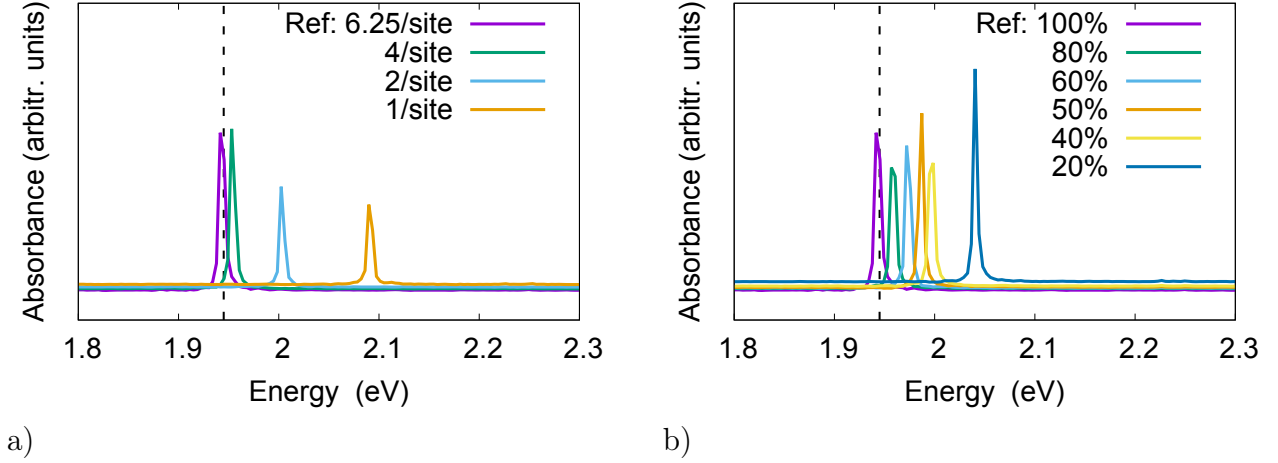


Figure 4: Active-space convergence of the defect-associated optical response for the 512-site NV- x 4 supercell at the Γ point, using transverse polarization along $[1\bar{1}0]$ direction. The dashed vertical line marks the experimental NV $^-$ zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The reference calculation, shown in both panels, uses the full occupied space, $f_{\text{occ}} = 100\%$, and $N_{\text{virt}} = 3200$, corresponding to $n_{\text{virt}} = 6.25$ virtual states per diamond site for the 512-site cell. (a) Dependence on the virtual active space, expressed as the number of virtual states per diamond site, $n_{\text{virt}} = N_{\text{virt}}/N_{\text{site}}$. The values $n_{\text{virt}} = 1, 2, 4$, and 6.25 correspond to $N_{\text{virt}} = 512, 1024, 2048$, and 3200 , respectively. Increasing n_{virt} systematically red-shifts the peak toward a saturated value. (b) Dependence on the retained occupied fraction, $f_{\text{occ}} = N_{\text{occ}}^{\text{act}}/N_{\text{occ}}^{\text{full}}$. The retained values 100%, 80%, 60%, 50%, 40%, and 20% are approximate and correspond to 1024, 800, 600, 512, 400, and 200 occupied states. Reducing f_{occ} shifts the defect-associated peak to higher energy.

than to moderate truncation of the occupied manifold. However, aggressive occupied-space truncation also produces a measurable blue shift, indicating that the response of deeper valence states contributes to the screening and relaxation of the localized defect excitation.

4.1.4 Defect concentration and magnetic alignment in replicated NV $^-$ arrays

The large NV- x 8 supercell also provides a controlled framework for examining how the optical response evolves with defect concentration. Starting from the same 4096-site diamond host, we constructed cells containing 1, 8, and 64 NV $^-$ centers. These systems correspond, respectively, to the defect concentrations obtained from the NV- x 8, NV- x 4, and NV- x 2 cells: the 8-NV structure is generated by a $2 \times 2 \times 2$ replication of the NV- x 4 cell, whereas the 64-NV structure is generated by a $4 \times 4 \times 4$ replication of the NV- x 2 cell. The comparison therefore changes the number and separation of the defects while retaining the same total supercell size, real-space grid, transverse field polarization, spectral broadening, and Γ -point representation. The concentration series discussed first uses a parallel collinear alignment of the NV moments.

The RT-TDDFT calculations for the three 4096-site systems also used the same active-space density. Approximately 67% of the occupied space was retained, while the virtual space contained approximately two unoccupied states per diamond site. In the RMG state-count convention, the calculations used 16400 computed states, with 2700 deep occupied states frozen and four highest-energy states discarded, leaving 13696 states in the density-matrix propagation. Each spectrum was obtained from 42000 electronic time steps with $\Delta t =$

1.0 a.u., corresponding to a total propagation time of approximately 1.02 ps. Consequently, differences among the three RT-TDDFT spectra predominantly reflect the NV concentration rather than changes in total cell size, propagation length, or active-space density.

For comparison, we also evaluated the optical response within the independent-particle approximation (IPA) using the LOPTICS implementation in RMG. In atomic units, the spin-resolved imaginary part of the IPA dielectric tensor can be written as

$$\epsilon_{\alpha\beta}^{(2),\sigma}(\omega) = \frac{4\pi^2}{\Omega\omega^2} \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{v,c} (f_{v\mathbf{k}\sigma} - f_{c\mathbf{k}\sigma}) \text{Re} \left[P_{v\mathbf{k}\sigma}^{\alpha*} P_{c\mathbf{k}\sigma}^{\beta} \right] \delta_{\eta} (\epsilon_{c\mathbf{k}\sigma} - \epsilon_{v\mathbf{k}\sigma} - \omega), \quad (96)$$

where Ω is the supercell volume, $w_{\mathbf{k}}$ is the $\mathbf{k}$ -point weight, and $P_{v\mathbf{k}\sigma}^{\alpha}$ is the velocity-gauge transition matrix element, including the nonlocal-pseudopotential contribution. The indices v and c denote occupied and unoccupied Kohn–Sham states of the same global spin σ , and δ_{η} represents the broadened energy-conservation condition. The IPA spectra therefore expose the underlying vertical Kohn–Sham transition structure, whereas the RT-TDDFT spectra additionally contain the self-consistent induced Hartree and adiabatic exchange–correlation response. The IPA curves in Fig. 5 were independently scaled for clarity, so the comparison concerns peak positions and spectral structure rather than absolute intensities.

Figure 5 shows a pronounced evolution of the optical response with decreasing NV concentration. For the 64-NV array, corresponding to the NV- $x2$ concentration, both IPA and RT-TDDFT produce strongly fragmented spectra over the sub-gap energy range. The large number of optically active features is consistent with substantial coupling among the closely spaced defects, which hybridizes the defect-derived states and produces dispersive occupied and unoccupied defect bands. The NV- $x2$ concentration is therefore clearly outside the isolated-defect regime.

At the intermediate NV- $x4$ concentration, represented by eight defects in the same 4096-site host, the spectral response becomes much simpler. The IPA spectrum retains several closely spaced transitions in the vicinity of the experimental ZPL energy, indicating that some fine structure remains in the underlying Kohn–Sham transition spectrum. By contrast, the self-consistent RT-TDDFT response is dominated by a single resonance near 2.01 eV. This simplification is consistent with coupling and redistribution of oscillator strength among nearby independent-particle transitions in the time-dependent response. Finite spectral resolution and the truncated RT-TDDFT active space may also suppress weaker components, so the single RT-TDDFT peak should not be interpreted as evidence that every periodic-image effect has disappeared. Nevertheless, it shows that the dominant optical observable is already substantially single-resonance-like at the NV- $x4$ defect separation.

Further dilution to a single NV^- center in the 4096-site NV- $x8$ cell produces a similarly simple low-energy response. The qualitative similarity of the dominant RT-TDDFT peaks for the 8-NV and 1-NV systems supports the conclusion that a separation corresponding to the 512-site NV- $x4$ model is sufficient to recover the principal defect-associated optical feature within the present spectral resolution and active-space approximation. The calculated peak near 2.01 eV remains approximately 0.06 eV above the experimental 1.945-eV ZPL. As shown by the active-space analysis in Sec. ??, part of this residual blueshift is consistent with the restricted occupied and virtual propagation spaces used for the 4096-site calculations. The experimental ZPL is nevertheless used only as a reference because the present frozen-ion

adiabatic-PBE response also omits excited-state structural relaxation and vibronic effects.

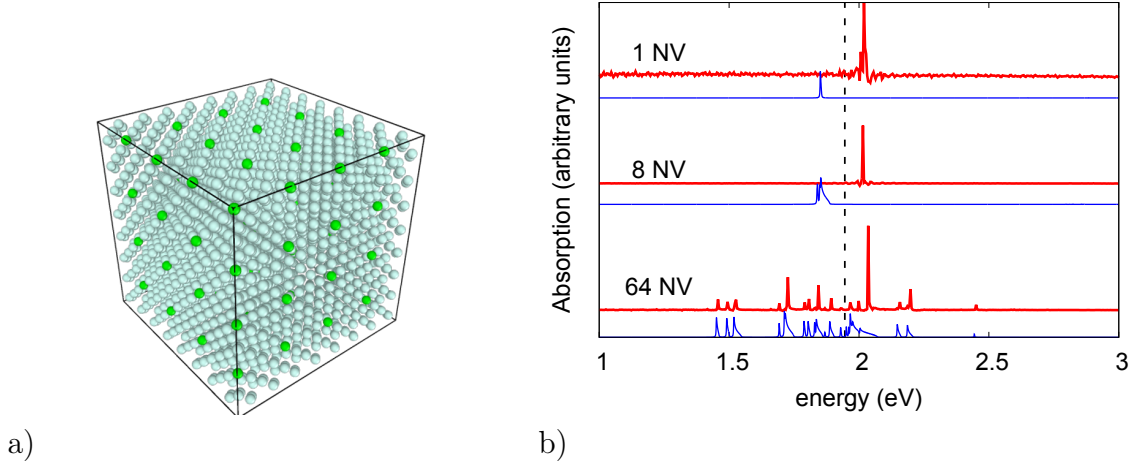


Figure 5: Concentration dependence of the optical response in parallel-aligned NV^- arrays. Spin-collinear RT-TDDFT spectra are shown in red and independent-particle approximation (IPA) spectra, evaluated with the LOPTICS implementation in RMG, are shown in blue. From top to bottom, the systems contain 1, 8, and 64 NV^- centers in the same 4096-site diamond host, corresponding to the NV- x 8, NV- x 4, and NV- x 2 defect concentrations, respectively. All calculations use the same atomic structures, real-space grid, Γ -point representation, transverse polarization, and spectral broadening. The RT-TDDFT calculations additionally use the same retained occupied fraction and number of virtual states per diamond site and were propagated for 42000 steps with $\Delta t = 1.0$ a.u., corresponding to approximately 1.02 ps. The dashed vertical line marks the experimental zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. The strongly fragmented spectrum of the 64-NV array reflects substantial coupling among closely spaced defects. At the 8-NV concentration, the IPA retains nearby independent-particle transition structure, whereas the RT-TDDFT response is dominated by a single resonance resembling that of the dilute single-NV system. The spectra are vertically offset, and the IPA curves are independently scaled, for clarity.

The fixed-volume replicated cells also permit examination of how the magnetic arrangement of the NV moments affects the spin-resolved optical response. We considered both a parallel collinear configuration, denoted FM below, and a G-type antiparallel broken-symmetry configuration, denoted G-AFM, for the 64-NV and 8-NV arrays. In the FM configuration, all local defect moments are aligned with the same global spin axis. In the G-AFM configuration, the local moment orientation alternates between neighboring defect sites.

The persistence of the local moments in these configurations can be characterized using the total and absolute magnetizations,

$$M_{\text{tot}} = \mu_B \int [\rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})] d\mathbf{r}, \quad M_{\text{abs}} = \mu_B \int |\rho_{\uparrow}(\mathbf{r}) - \rho_{\downarrow}(\mathbf{r})| d\mathbf{r}. \quad (97)$$

For the 8-NV array, the FM solution gives $M_{\text{tot}} = 20.73 \mu_B$ and $M_{\text{abs}} = 22.36 \mu_B$, whereas the G-AFM solution gives $M_{\text{tot}} = 0$ and $M_{\text{abs}} = 18.66 \mu_B$. The corresponding values for the 64-NV array are $M_{\text{tot}} = 128.42 \mu_B$ and $M_{\text{abs}} = 143.03 \mu_B$ for the FM arrangement, and $M_{\text{tot}} = 0$ and $M_{\text{abs}} = 138.57 \mu_B$ for the G-AFM arrangement. The vanishing total magnetization but substantial absolute magnetization of the G-AFM solutions confirms that the local defect moments remain present and cancel globally rather than disappearing.

The spin-resolved spectra in Fig. 6 show how the local minority-spin selection rule established for the single NV center maps onto the globally defined spin channels of a multi-defect array. In the FM configuration, all local moments have the same orientation. The local minority-spin $a_1 \rightarrow e$ excitation therefore corresponds to the same global spin channel at every defect, and the optical response is strongly dominated by one spin component. In the G-AFM arrangement, the local moment is reversed on the second magnetic sublattice. The local minority-spin excitation consequently maps onto one global spin channel on one sublattice and the opposite global spin channel on the other. The corresponding defect-associated features therefore appear in both global spin-resolved spectra.

This redistribution between global spin channels does not represent a spin-flip optical transition. The optical perturbation remains spin independent and the transitions conserve the global collinear spin label. Instead, the result reflects the reversal of the local moment relative to the fixed global quantization axis. Local minority spin is represented by opposite global spin labels on the two G-AFM sublattices.

The concentration dependence found above is also evident in the magnetic comparison. At the NV- $x2$ concentration, both FM and G-AFM spectra contain many defect-associated peaks, and the detailed distribution of these features depends on the magnetic arrangement. At the more dilute NV- $x4$ concentration, the response is concentrated primarily in a single resonance near 2.0 eV. Changing from FM to G-AFM then affects mainly the distribution of this resonance between the two global spin channels rather than producing a comparably strong fragmentation of the spectrum. This behavior is consistent with a substantial reduction of electronic coupling among the NV centers as their separation increases.

Together, these calculations show that increasing the NV–NV separation transforms the optical response from the fragmented spectrum of an interacting defect array to a dominant isolated-defect-like resonance. At the same time, the magnetic alignment determines how the local minority-spin excitation is represented in the globally defined spin channels. The 4096-site calculations therefore demonstrate that the periodic spin-collinear RT-TDDFT implementation can treat both dilute defects and extended arrays containing as many as 64 optically active defect centers within the same real-space simulation framework.

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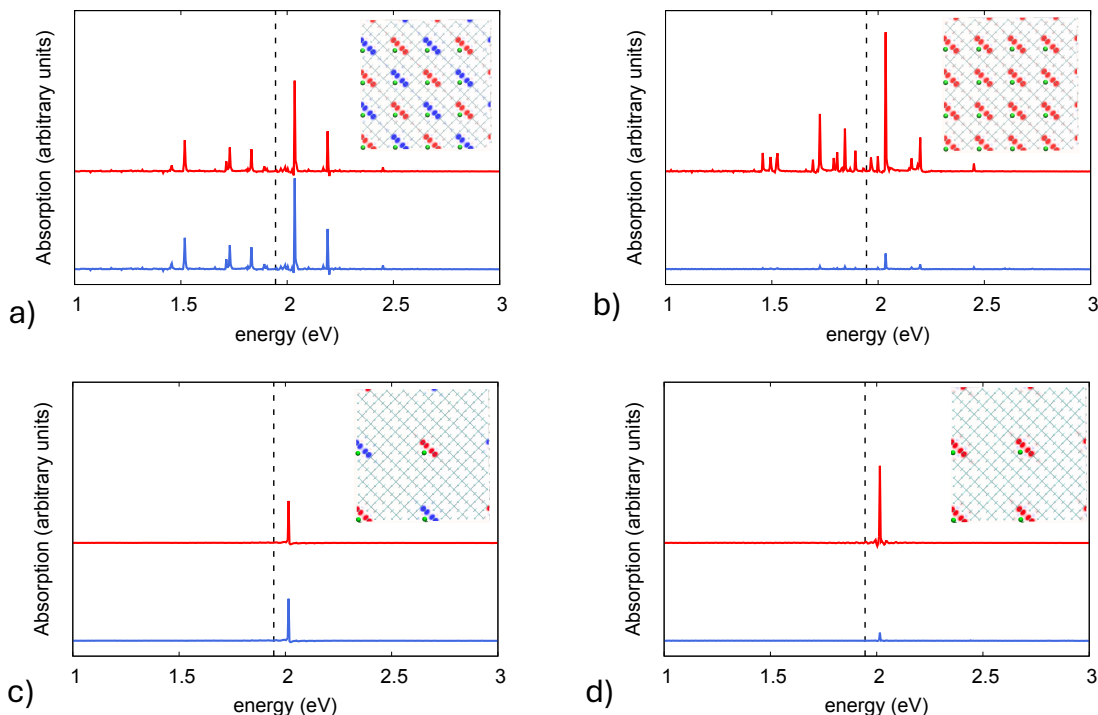


Figure 6: Spin-resolved RT-TDDFT optical response of replicated NV^- arrays in the 4096-site diamond host. The upper row shows the 64-NV system at the $\text{NV-}x2$ concentration, and the lower row shows the 8-NV system at the $\text{NV-}x4$ concentration. Panels (a) and (c) correspond to G-type antiparallel broken-symmetry arrangements, whereas panels (b) and (d) correspond to parallel, FM arrangements. The red and blue curves show the two globally defined collinear spin channels and are vertically offset for clarity. The insets illustrate the corresponding local defect-spin arrangements; red and blue markers denote opposite local moment orientations. The dashed vertical lines mark the experimental NV^- zero-phonon-line energy, $E_{\text{ZPL}} = 1.945$ eV. In the FM configurations, the local minority-spin defect excitation maps predominantly onto one global spin channel. In the G-AFM configurations, reversal of the local moments between the two magnetic sublattices causes the same local minority-spin excitation to appear in both global spin channels. The response remains strongly fragmented at the $\text{NV-}x2$ concentration but becomes predominantly single-resonance-like at the $\text{NV-}x4$ concentration.

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